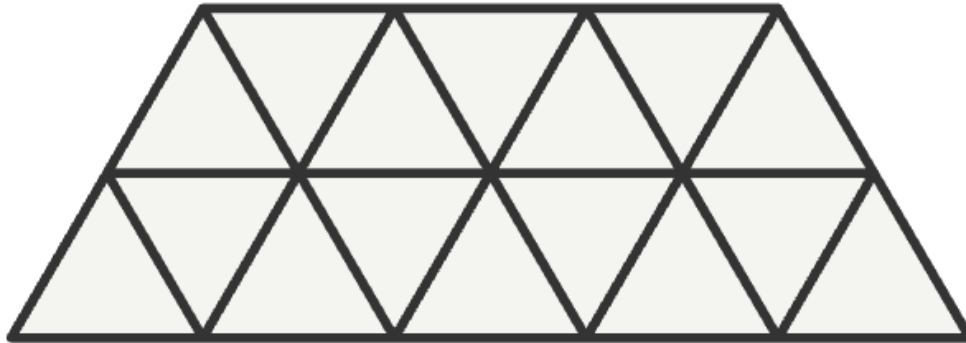
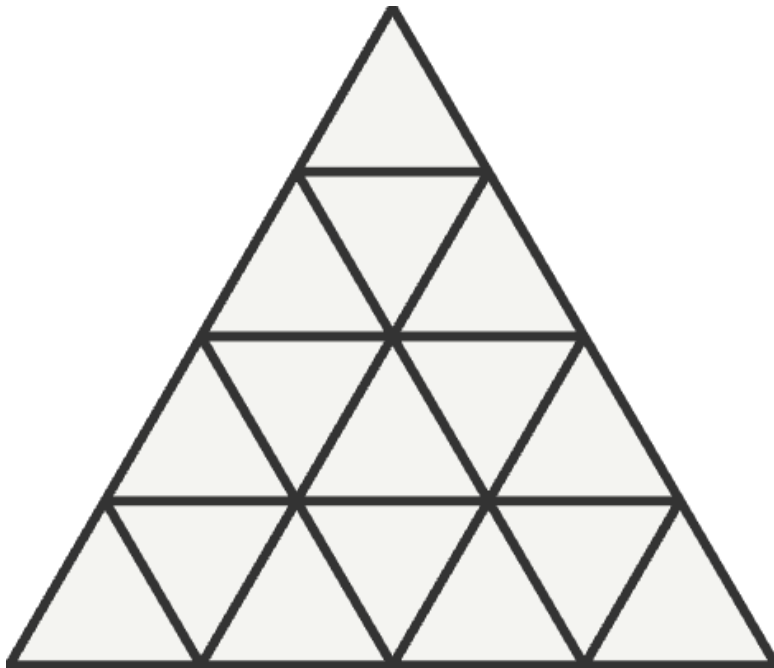


Pattern Block Exercises

Problem 1: Tile the following shape with one each of the green, purple, yellow, blue, and red pieces. (The paper might not be exactly the right size - the small green triangle is supposed to be the same size as the triangles on the paper.)



Problem 2: If I want to fill up this shape entirely by using the red trapezoid tile, how many trapezoids will I need? How many triangles will be left uncovered? NOTE: we're not allowed to have any trapezoids go over the edge.



Problem 3: Is it possible for any of the triangles to be the one left over? Which ones don't work?

Problem 4: Because of symmetry, we don't need to check every triangle. For example, the corners are all the same "type" of triangle. How many "types" of triangle are there total? Label them with the letters A, B, C, etc.

Fun fact: mathematicians call these "types" of things **equivalence classes**. So another way to ask this question is: How many equivalence classes of triangles are there?